

Bright source simulations

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Outline

We've prepared several simulation scripts which can be run and modified.

They are broadly categorized into

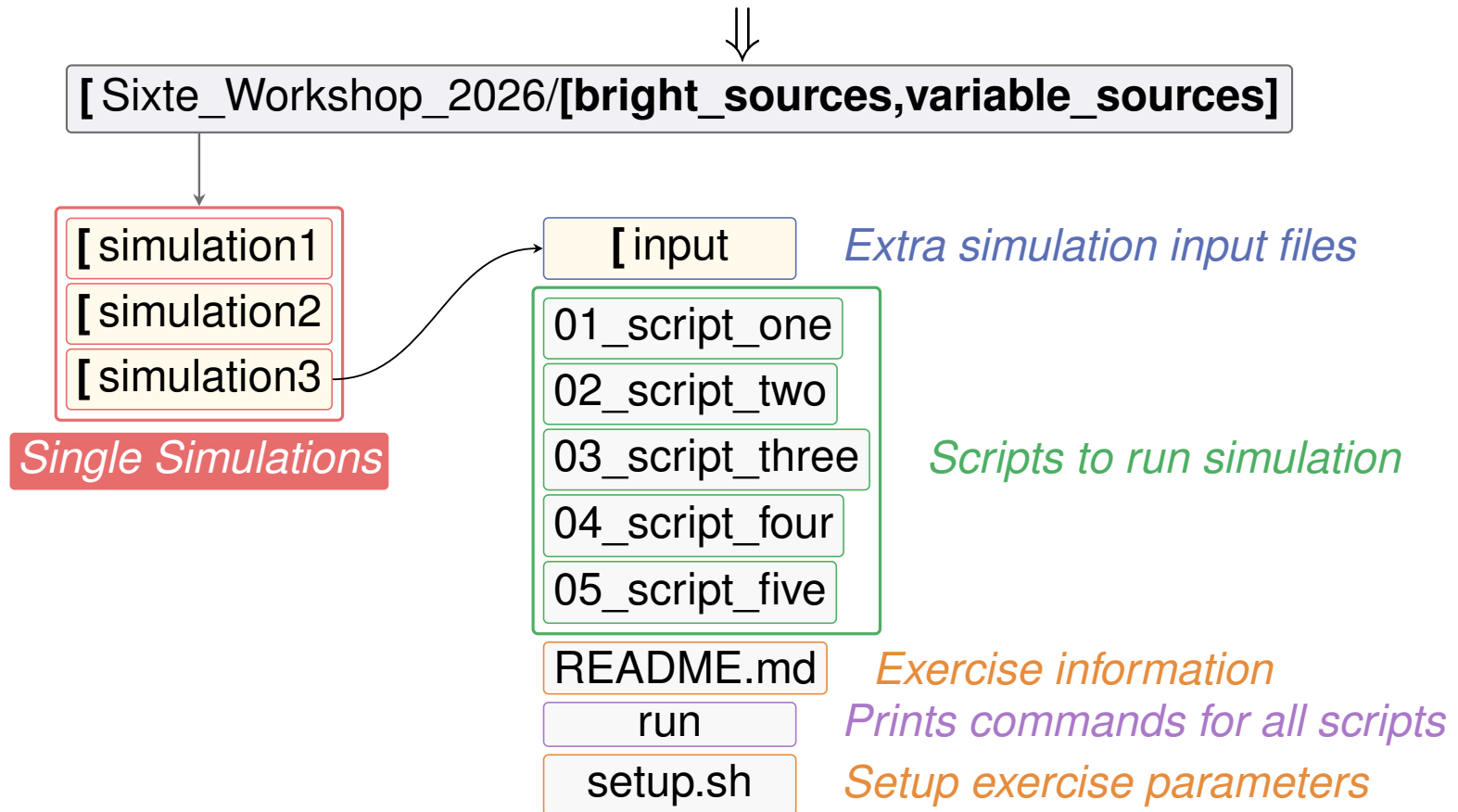
Bright Sources Examples that analyze important detector effects for the WFI (pile-up) and the X-IFU (branching ratios)

Variable Sources Creating and analyzing variable sources (light curves, PSDs, ...)

Detailed help and tutorials on all Sixte Tools are in the **Simulator Manual** (<https://www.sternwarte.uni-erlangen.de/sixte/sixte-manual/>)

Where to find the files: GitHub Repository

```
git clone https://github.com/stierhofj/Sixte_Workshop_2026.git
```



Simulation Examples

Bright Sources

[simple_src_xifu](#) basic simulations to understand grading and the branching ratios

[microcalorimeter](#) building upon the simple example, also extract spectra for different grades

[pileup](#) basic simulations to understand pileup with the WFI

[python_exercise](#) longer tutorial in exercise style to show how python can be used to create a SIMPUT for multiple sources, and analyze pile-up

Variable Sources

[periodic_lc](#) simple example that shows how to build a simput for a source with a periodic light curve

[psd](#) simple example that shows how to build a simput give a PSD

Bright Sources with *NewAthena*

(see the *WFI* and *X-IFU* introduction talks for the flux limits)

Important to choose the right *observation mode*: **defocused optics**

WFI: `fd_wfi_ff_df35mm.xml`

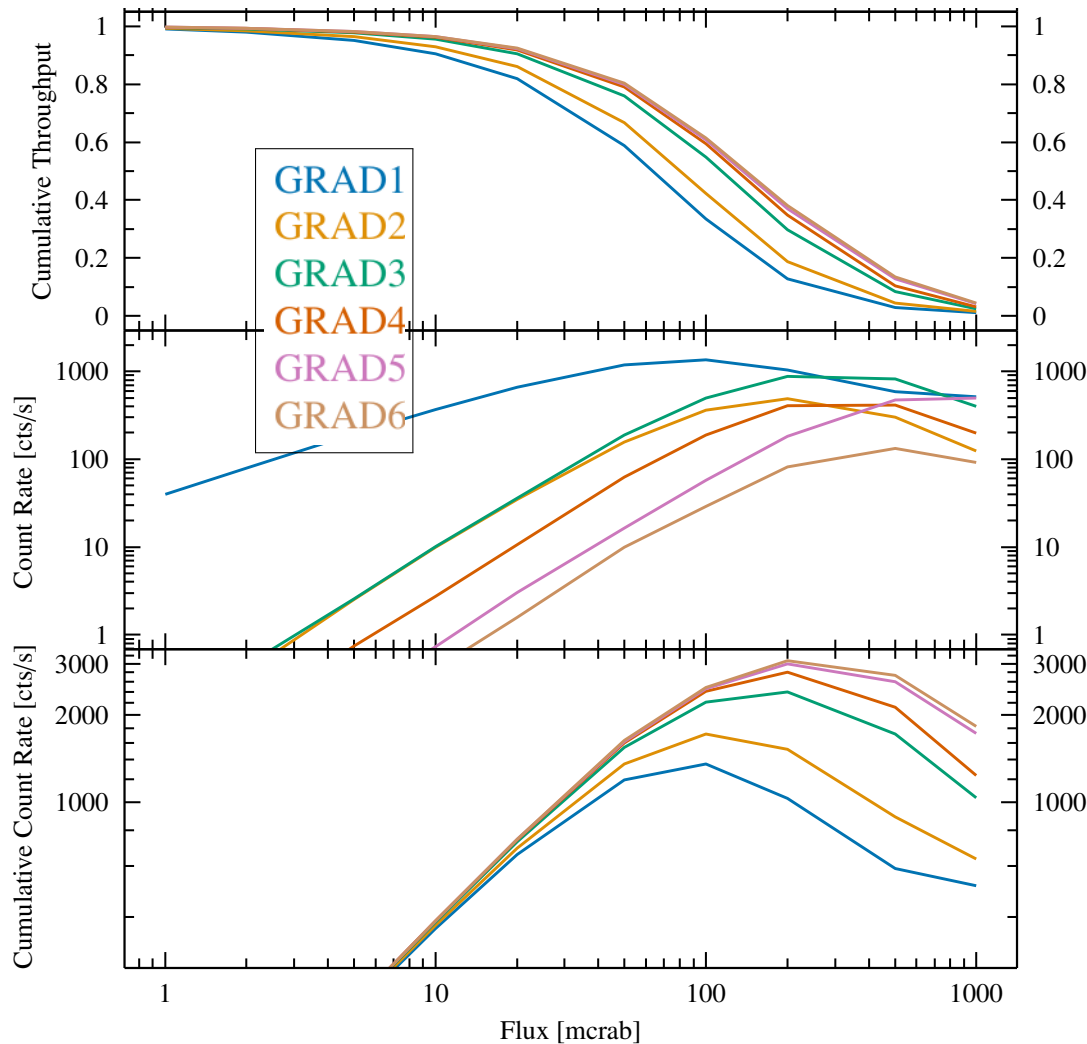
X-IFU: `xifu_optical_filter_defoc.xml`

For the brightest sources: use thick filter (removes all photons below 2 keV)

WFI: `fd_wfi_ff_df35mm_thick_filter.xml`

X-IFU: `xifu_be_filter_defoc.xml`

Observing Mode for X-IFU: Defocused incl. Optical Filter



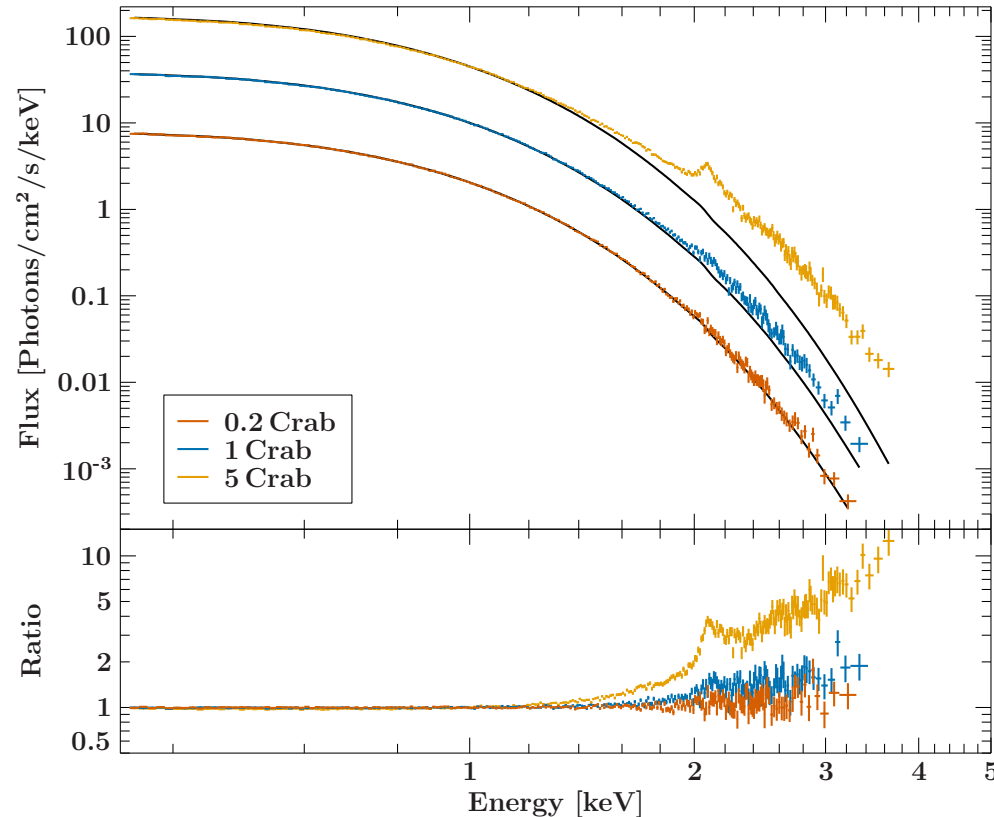
High resolution 4 eV
(GRAD1 & GRAD2) events
strongly suppressed
> 50 mCrab

⇒ use conservative approach to extract everything as **Intermediate Resolution 4.2 eV** (GRAD3) events

Cumulative Throughput/Counts
means for a given flux, throughput/Counts are added for the given GRADE X and better (meaning $\text{GRADE} \leq \text{GRADE X}$)

for bright 50-200 mCrab sources, intermediate resolution (4.2 eV) is ideal

The WFI and pile-up



pile-up distorts the spectral shape

the limits for the WFI are
(see presentation)

defocus XYZ

defocus + Be filter XYZ

ways to check if pile-up is a problem in simulations

1. verify that pile-up fraction is below 1% (see WFI presentation and Dauser et al., 2019 for more details), and exercise `pileup`
2. test that the simulated spectrum matches the input spectral shape